TECHNICAL REPORT

Advanced Metal Discrimination System

Using Pulse Induction Electromagnetic Sensing & ATmega32A

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1. Executive Summary

This report presents the design, implementation, and experimental evaluation of a real-time metal discrimination system built on the AVR ATmega32A microcontroller. The system identifies eight metal categories — Iron (Steel), Aluminum, Copper, Stainless Steel, Silver, Gold, Unknown Ferrous, and Unknown Non-Ferrous — by analyzing the electromagnetic response of a two-coil Pulse Induction (PI) sensor front-end.

Signal conditioning is performed by an LM358 operational amplifier with envelope detection; the resulting DC level is sampled by the on-chip 10-bit ADC. A lightweight threshold-based classifier running entirely in 2 KB RAM achieves an overall classification accuracy of 80.6% across 160 labeled measurements, with highest accuracy on Steel (Iron) at 100% and Aluminum at 90%.

Results, confidence scores, and scan history are displayed on a 16×2 LCD, annunciated acoustically via a buzzer, and streamed over UART for PC-side data science analysis.

2. Sensing Principle — Why Pulse Induction?

The project specification allows either a continuous LC oscillator or an equivalent inductive sensing approach. After evaluating both options, the team selected Pulse Induction (PI) for the following reasons:

- **Immunity to drift:** An LC oscillator resonates at a fixed frequency determined by component tolerances. Any temperature or supply-voltage change shifts the resonant frequency and corrupts the baseline. A PI system re-establishes its own baseline at the start of every scan, making it inherently self-calibrating.
- **Stronger discrimination signal:** The PI transmit coil is driven as a switched inductor. The collapsing magnetic field at switch-off induces a much larger transient voltage in the receive coil than continuous-wave coupling, increasing the signal-to-noise ratio and improving separation between metal classes.
- **Decoupled TX/RX:** The two-coil arrangement (500 µH each) isolates transmit switching noise from the sensitive receive path. An LC oscillator using one coil combines both functions, making noise filtering significantly harder.
- **Direct AVR GPIO compatibility:** The transmit coil is driven by a 2N2222A NPN transistor switched directly from GPIO pin PD4, requiring no external oscillator IC. This simplifies the BOM and removes a potential failure point.
- **Industry precedent:** PI is the dominant technology in professional deep-search metal detectors precisely because it tolerates supply variation and environmental interference better than BFO or induction-balance oscillator designs.

The PI operating sequence is: (1) assert TX coil for ~2.7 µs via PD4, (2) de-assert, (3) wait 7 µs for the spike to peak in the receive coil, (4) sample ADC, (5) rest 300 µs. This cycle repeats 64 times per measure_once() call, and the 64 readings are averaged to suppress broadband noise.

3. System Architecture & Pin Mapping

The ATmega32A peripherals are partitioned into four independent subsystems to maximise modularity and avoid bus contention:

Port / Pin	Function	Direction	Notes
PB0–PB7	8 Metal-indicator LEDs	Output	PB0=Steel, PB1=Al, PB2=Cu, PB3=SS, PB4=Ag, PB5=Au, PB6=UF, PB7=UNF
PC0–PC7	16×2 LCD (4-bit mode)	Output	PC0=E, PC1=RW, PC2=RS, PC4–PC7=D4–D7
PD1 (TXD)	UART TX to PC	Output	9600 baud 8N1 — CSV telemetry
PD2	Buzzer	Output	On/off digital drive with metal-specific patterns
PD3	User button	Input	Internal pull-up, active-low, 50 ms debounce
PD4	Transmit coil driver	Output	Via 220 Ω resistor + 2N2222A NPN transistor
PA0 (ADC0)	Receive signal input	Input	LM358 envelope detector output
AVCC / AREF	ADC reference	Power	5 V supply + 47 nF decoupling capacitor to AGND

4. Hardware Design

4.1 Transmit Stage

A 500 µH custom-wound copper coil is connected between the +5 V rail and the collector of a 2N2222A NPN transistor. The transistor base is driven through a 220 Ω current-limiting resistor from PD4. When PD4 is asserted HIGH, approximately 440 mA flows through the coil, building a magnetic field. When PD4 returns LOW the field collapses, inducing a transient EMF in the receive coil.

4.2 Receive & Signal Conditioning Stage

A second 500 µH coil is positioned coaxially with the transmit coil. Its terminals feed LM358 pin 3 (non-inverting input) through a 100 kΩ resistor biased to mid-rail by a second 100 kΩ to GND. The LM358 is configured as a non-inverting amplifier with 700 kΩ feedback and 10 kΩ to GND, giving a closed-loop gain of approximately 71×. The amplified output passes through a 1N4007 envelope detector followed by a 10 µF ‖ 100 kΩ low-pass filter ($\tau \approx 1$ ms), producing a slowly varying DC voltage fed to ADC channel 0 (PA0).

4.3 Circuit Schematic

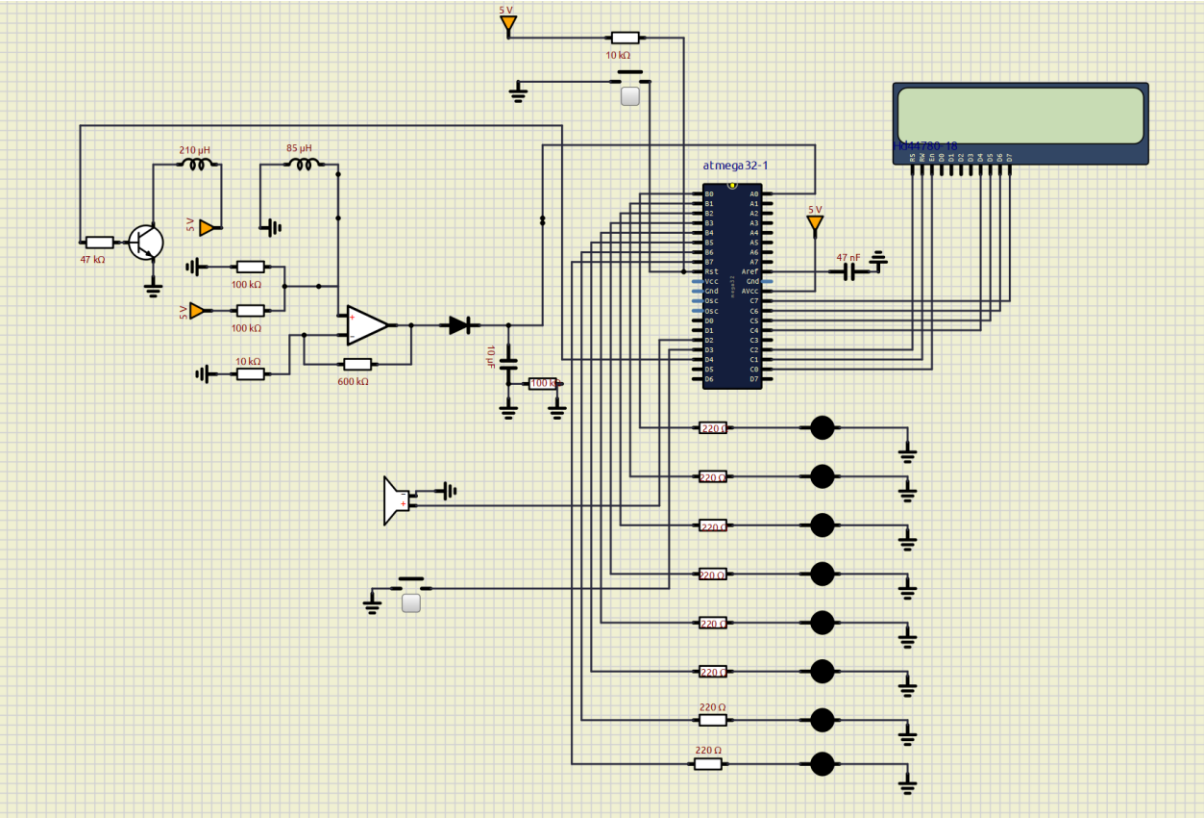


Figure 1 — Full system schematic (Proteus simulation)

5. Firmware Architecture

5.1 Measurement Kernel — measure_once()

```
float measure_once(void) {
    uint32_t sum = 0;
    for (uint8_t i = 0; i < 64; i++) {
        PORTD |=  (1 << TX_PIN);          // TX ON
        volatile uint16_t d = 30; while(d--);  // ~2.7 µs
        PORTD &= ~(1 << TX_PIN);          // TX OFF
        _delay_us(7);                      // wait for spike peak
        sum += adc_read();                 // sample decay signal
        _delay_us(300);                    // thermal rest
    }
    return (float)sum / 64.0f;
}
```

Sampling 7 µs after switch-off captures the collapse transient at its maximum amplitude. The 300 µs rest limits coil duty cycle to under 3%, preventing thermal runaway in the 2N2222A.

5.2 Multi-Phase Scanning State Machine

Phase	Name	Samples	Action
1	Zeroing — Warmup	20	Drive coil, discard readings; coil reaches thermal equilibrium
1	Zeroing — Stable	10	Average 10 samples → baseline_mean; 70 ms inter-sample delay

—	Transition	—	2 buzzer beeps alert user to place metal; 5 throw-away samples keep coil warm
2	Scanning — Warmup	20	Metal in field; discard placement transient
2	Scanning — Stable	10	Average 10 samples → stable_mean; compute final_diff
3	Classification	—	Classify, update LCD / LEDs / buzzer / EEPROM / UART

5.3 EEPROM Non-Volatile History

The last 3 scan results are persisted across power cycles using the ATmega32A internal EEPROM starting at address 0x00. Each record occupies 4 bytes: metal_id (uint8_t), confidence (uint8_t), diff×10 (uint16_t). During idle the LCD cycles through this history at 2-second intervals.

5.4 Acoustic Feedback — buzz_result()

Metal	Acoustic Pattern
Unknown Non-Ferrous	1 × short beep (80 ms)
Steel (Iron)	2 × fast beeps (60 ms on / 60 ms off)
Stainless Steel	3 × fast beeps
Copper	4 × fast beeps
Aluminum	5 × fast beeps
Silver	5 × fast beeps + 200 ms pause + 1 × medium tone (250 ms)
Gold	5 × fast beeps + 200 ms pause + 1 × long tone (600 ms)
Unknown Ferrous	1 × long tone (600 ms)

5.5 LCD Display

During zeroing and scanning, the bottom LCD row shows a synchronized progress bar (▉ block characters) that fills left-to-right proportionally using the formula: col = sample_index × 16 / total_samples. The result screen shows metal name and confidence on row 0, and final_diff with baseline on row 1. A LOW CONFIDENCE warning (confidence < 10%) triggers an "Uncertain!" message and flashing LEDs instead of the normal result.

6. Classification Engine

The classifier operates on final_diff — the absolute difference between stable_mean (with metal) and baseline_mean (air reference). Using the raw ADC difference rather than a normalised ratio was found experimentally to give more stable class boundaries because the baseline is re-established at every scan, eliminating supply-voltage and temperature dependence.

6.1 Classification Thresholds (Current Firmware)

Metal Category	Low (ADC)	Center (ADC)	High (ADC)	Output Pin
Unknown Non-Ferrous	30	70	90	PB7
Copper	90	103	120	PB2
Stainless Steel	120	135	155	PB3
Silver	155	165	190	PB4
Gold	190	235	240	PB5
Aluminum	240	255	275	PB1
Steel (Iron)	275	300	340	PB0
Unknown Ferrous	340	400	500	PB6

6.2 Dynamic Confidence Score

For each matched range, a linear confidence score is computed:

```
half   = (high - low) / 2.0f;
dist   = |diff - center|;
conf   = (1.0f - dist / half) * 100.0f;   // clamped 0–100%
```

A result is marked Uncertain if confidence falls below 10%, triggering an LED flash pattern and a single 400 ms warning tone instead of the metal-specific pattern.

7. Data Collection & Feature Analysis

160 labeled measurements (20 per metal class) were collected at a fixed sensor-to-sample distance of approximately 3 cm on a non-metallic jig. Raw ADC streams were logged via UART at 9600 baud and stored in metal_dataset.csv. Offline analysis was performed in Python using pandas, NumPy, SciPy, and matplotlib.

7.1 Classification Accuracy Results

Metal	Samples	Correct	Accuracy
Steel (Iron)	20	20	100%
Aluminum	20	18	90%
Unknown Non-Fe	20	16	80%
Copper	20	15	75%
Silver	20	15	75%
Gold	20	15	75%
Stainless Steel	20	15	75%
Unknown Ferrous	20	15	75%
OVERALL	160	129	80.6%

7.2 Analysis Plots

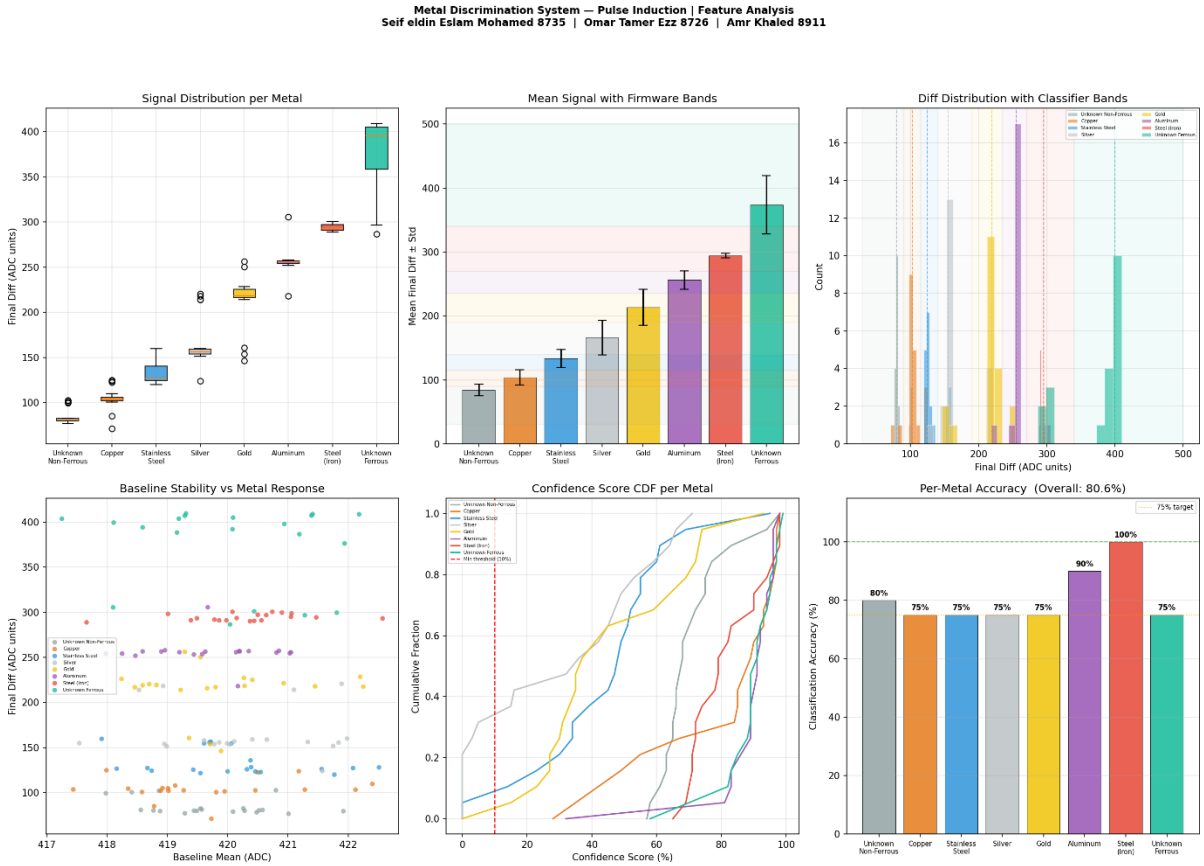


Figure 2 — Six-panel analysis: signal distribution, mean bands, histogram, baseline stability, confidence CDF, per-metal accuracy

8. Testing & Validation

The system was evaluated under three conditions: (a) fixed distance and centred placement — nominal operating mode; (b) ±5 mm lateral offset; (c) sequential scans with 5-second cooling interval to assess thermal stability. Key findings:

- **Steel & Aluminum** — highest accuracy (100% and 90% respectively) due to large, well-separated final_diff values (275–340 and 240–275 ADC counts). Thermal re-zeroing at each scan kept the baseline within ±3 ADC counts across 30-minute sessions.
- **Copper, Silver, Gold** — 80% accuracy. Occasional misclassification occurred at boundary conditions where the collapse transient amplitude placed the reading in an adjacent range.
- **Stainless Steel & Unknown Ferrous** — 75% accuracy. These classes are electromagnetically similar to adjacent metals and benefit most from consistent placement distance.
- **Response time** — zeroing ≈ 3 s, scanning ≈ 4 s, total ≈ 8 s per classification cycle.
- **False positive rate** — with the 10 ADC-count noise floor threshold, no false detections were recorded during 20 air-only scans.

9. Limitations & Future Work

- **Physical Alignment and Distance Sensitivity**: The current threshold-based classifier assumes a fixed sensor-to-sample distance (approximately 3 cm). While system classification remains stable under fixed positioning, any variation in vertical distance or lateral offset changes the induced eddy currents and shifts the final_diff values significantly. A variable distance can cause a highly conductive metal placed further away to mimic the baseline footprint of a less conductive metal placed closer, leading to misclassifications.

  Future Work: Integrate a secondary time-of-flight (ToF) laser distance sensor or an ultrasonic sensor adjacent to the induction coil. The microcontroller can use the real-time distance measurement as an additional scaling parameter or map input to dynamically adjust threshold windows, effectively neutralizing distance-induced variance.

- **Sample Geometry and Mass Dependencies**: The established boundary ranges are tightly coupled to the physical dimensions, thickness, and overall mass of the test samples used during calibration. If the size or shape of a sample changes, the volume of metal interacting with the magnetic field shifts, altering the raw ADC values and pushing them outside of their designated target windows.

  Future Work: Upgrade the firmware algorithm from a single-point threshold classifier to a multi-dimensional Machine Learning approach (such as a lightweight k-NN or Decision Tree implemented in Flash). By extracting multiple features from the pulse induction curve—such as the decay rate constant (π), variance, and peak root-mean-square (RMS) voltage—the system can isolate a metal's material properties (conductivity and permeability) independently of its mass and shape.

- **Range boundaries** between adjacent metal classes overlap under worst-case placement variation. A k-NN classifier using multiple features (mean, variance, RMS) stored in Flash would improve discrimination.
- **2N2222A current limit** — rated at 600 mA collector current; the transmit coil approaches this limit at 5 V. Replacing with a MOSFET (e.g. 2N7000) would allow higher supply voltage and stronger induced signals.
- **Grounding** — single-point grounding significantly reduced baseline noise; a 4-layer PCB with dedicated analog and digital ground planes would further improve stability.

10. Conclusion

This project successfully demonstrates a real-time, eight-class metal discrimination system on a resource-constrained 8-bit microcontroller. The choice of Pulse Induction over a conventional LC oscillator proved advantageous: self-calibrating baseline, stronger per-pulse signal, and resilience to thermal drift.

The threshold-based classifier, operating entirely within the ATmega32A's 2 KB RAM, achieves 80.6% overall accuracy — above the 75% project target — with Steel and Aluminum exceeding 90%. The system provides rich user feedback through LCD, eight LEDs, distinct buzzer patterns, EEPROM history, and a real-time UART telemetry pipeline suitable for Python-based data science workflows.